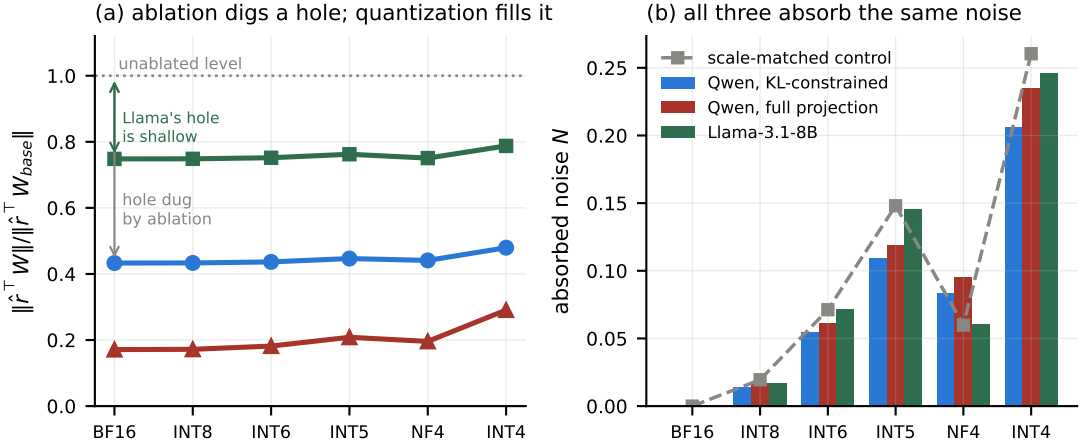
(a) ablation digs a hole; quantization fills it
unablated level
Llama's hole
is shallow
hole dug
by ablation
$\|\hat{r}^\top W\|/\|\hat{r}^\top W_{base}\|$
1.0
0.8
0.6
0.4
0.2
0.0
BF16
INT8
INT6
INT5
NF4
INT4
(b) all three absorb the same noise
scale-matched control
Qwen, KL-constrained
Qwen, full projection
Llama-3.1-8B
absorbed noise N
0.25
0.20
0.15
0.10
0.05
0.00
BF16
INT8
INT6
INT5
NF4
INT4